



RAJIV GANDHI UNIVERSITY OF KNOWLEDGE TECHNOLOGIES, BASAR
Civil Engineering
(A.Y. 2025-2026)

Subject Name: FM
Exam: E2/S1/MT1
Date: 08/08/2025

Sub code: CE2104
Max. Marks: 15
Time: 45 min

B220367

SECTION -A (5*1=5M) ; Answer all the questions

1. A mercury water manometer has a gauge difference of 0.8m. The difference in pressure measured in meters of water is -----
2. For the immersion of vertical plane surface, the location of centre of pressure is -----the location of centre of gravity.
3. The movement of two blocks of wood weltd with hot glue requires greater and greater effort as the glue is drying up. Mention is whether it is true or false.
4. Write the dimensions of coefficient of viscosity.
5. Write the relation between mass density and weight density.

SECTION-B (2*5=10M) ; Answer any two questions

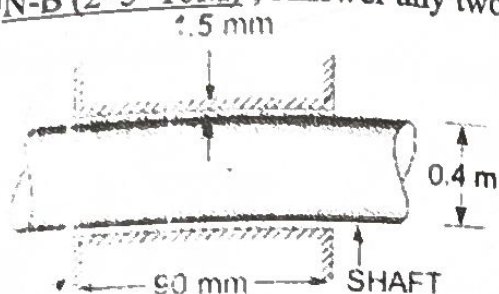


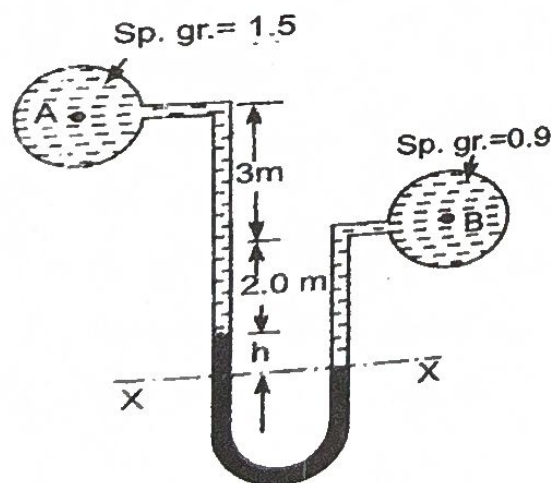
Figure 1

1. The dynamic viscosity of oil, used for lubrication between a shaft and sleeve is 6 poise. The shaft is of diameter 0.4m and rotates at 190 r.p.m. Calculate the power lost in the bearing for a sleeve length of 90mm. The thickness of the oil film is 1.5mm. [5M]

2. State and Prove Pascal's law.

Calculate the surface tension in a soap bubble of 50mm diameter with its inside pressure being 2.5N/m^2 above the atmospheric pressure. [3+2M]

3. A differential manometer is connected at the two points A and B of two pipes as shown in Figure beside. The pipe A contains a liquid of specific gravity 1.5 while the pipe B contains a liquid of specific gravity 0.9. The pressure at A and B are 98.1kPa and 176.58kPa respectively.



Find the difference in mercury level in the differential manometer.

Derive an expression for total pressure and point of application of total pressure for a vertically immersed plane surface. [2.5+2.5M]



Subject: Fluid Mechanics
Date: 24/09/2025
Exam: E2S1MT2

Student ID: 8220367

Subject Code: CE2104
Max Marks: 15
Time: 45 Min

SECTION -A (5*1=5M)

- 1) If $u = -3x - 3y + 2z$ and $v = 2x + 3y - 7z$ Then whether the flow is possible or not,
- 2) If $v = 3x - 7y + 3z$ $u = 5x - 6y + z$ and $w = 2x + 3y + kz$ then the value of 'K' for flow to be possible
- 3) Match the following

List-1

- (i) Fluid static
- (ii) Fluid kinematics
- (iii) Fluid Dynamics

List-2

- (1) Study of fluid in motion along with forces causing the motion
- (2) Study of fluid at rest
- (3) Study of fluid in motion
- (4) Study of forces causing the motion

Codes: (i) (ii) (iii)

(a) 3 2 1

(b) 2 3 4

(i) (ii) (iii)

(c) 2 3 1

(d) 2 1 3

- 4) If the discharge in two circular orifices O_1 and O_2 placed at a depth of 200mm and 800mm in water are equal, the ratio of diameter is...
- 5) Rotation of a fluid particle is caused by viscosity. Hence in inviscid region, flow is irrotational. [True/False]

SECTION -B (2*5=10M)

- 1) State Steady and uniform flow, Derive Eulers equation of motion. [5m]
- 2) Solve the following? Problem 5.16 The velocity components in a two-dimensional flow field for an incompressible fluid are as follows: $u = (y^3)/3 + 2x - x^2y$ and $v = xy^2 - 2y - x^3/3$
Obtain an expression for the stream function . [5m]
- 3) Define and elaborate the following
Bernoullis Equation, Laminar and Turbulent flow, Define-Equations of Stream function and Velocity Potential [5M]

-----THE END-----

P.T.O



Subject: Fluid Mechanics

Date: 30/10/2025

Exam: E2S1MT3

Subject Code: CE2104

Max Marks: 15

Time: 45 Min

SECTION -A (5*1=5M)

- 1) In case of laminar flow, the loss of pressure head is proportional to.....
- 2) The difference between total energy line and the hydraulic gradient line is equal to.....
- 3) A loss of head at a sudden contraction in a pipe is smaller than that at a sudden expansion. (True/False)
- 4) The shear in turbulent flow is mainly due to.....
- 5) Two long pipes in parallel are used to carry water between two reservoirs. The diameter of one pipe is twice that of the other. Both the pipes have the same value of friction factor. Neglect minor losses. What is the ratio of flow rates through the two pipes?

SECTION -B (2*5=10M)

- 1) State Different losses in pipes. Derive an expression for loss of head due to friction-Darcy weisbach equation? [5m]
- 2) A lubricating oil of viscosity 1 poise and specific gravity 0.9 is pumped through a 30 mm diameter pipe. If the pressure drop per metre length of pipe is 20 kN/m^2 , determine:
 - (i) The mass flow rate in kg/min,
 - (ii) The shear stress at the pipe wall.
 - (iii) The Reynolds number of flow and
 - (iv) The power required per 50 m length of the pipe to maintain the flow. [5m]
- 3) A horizontal pipe 150 mm in diameter is joined by sudden enlargement to a 225 mm diameter pipe. Water is flowing through it at the rate of $0.05 \text{ m}^3/\text{s}$, Find
 - (i) Loss of head due to abrupt expansion,
 - (ii) Pressure difference in the two pipes, and
 - (iii) Change in pressure if the change of section is gradual without any loss. [5M]

-----THE END-----



Subject Name: FLUID MECHANICS

Date: 24/11/25

Exam: E2/S1/REG-I/NOV

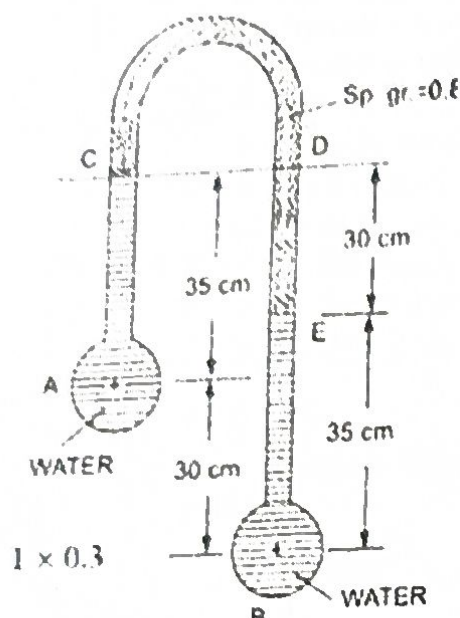
SubCode: CE2104

Max Marks: 60

Time: 3 Hours

ANSWER ANY 5 QUESTIONS FROM THE FOLLOWING

Note: Draw the neat diagrams where ever required.



1. A. Draw a neat sketch showing the types of fluids and explain in detail.

B. Determine the difference of pressure between the pipes in the figure [6+6M]

2.A. A wooden block of specific gravity 0.7 and having a size of 2m x 0.5m x 0.25 m is floating in water. Determine the volume of concrete of specific weight 25kN / (m³) = that may be placed which will immerse (i) the block completely in water, and (ii) the block and concrete completely in water.

B. A trapezoidal 2 m wide at the bottom, 4m wide at top and 1 m deep has side slopes 1: 1. Determine: (i) Total pressure;

(ii) Centre of pressure on the vertical gate closing the channel when it is full of water. [6+6M]

full of water.

3. Derive three dimensional continuity equation in Cartesian coordinates

The velocity components in a two-dimensional flow field for an incompressible fluid are as follows: $u = y^3/3 + 2x - x^2y$ and $v = xy^2 - 2y - x^3/3$

Obtain an expression for the stream function ψ .

[12M]

4. Write a note on classification of Notches and weirs.

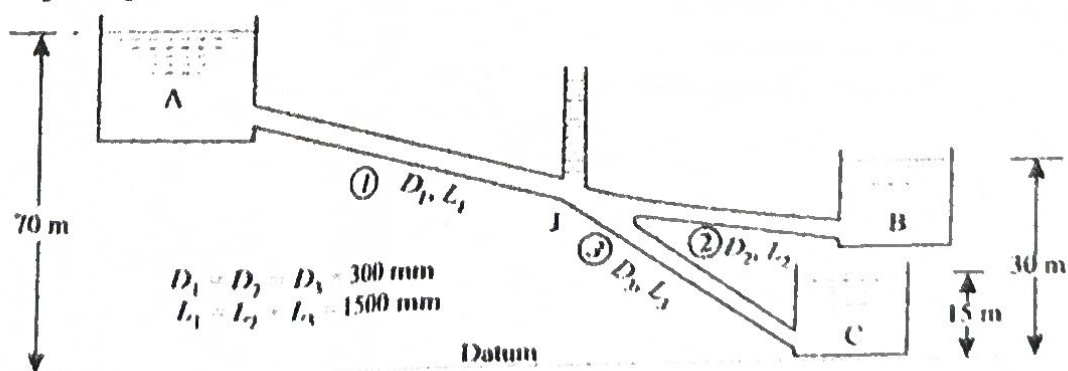
In a 45° bend a rectangular air duct of 1 m² cross-sectional area is gradually reduced to 0.5 m² area. Find the magnitude and direction of force required to hold the duct in position if the velocity of flow at 1 m² section is 10 m/s, and pressure is 30 kN/m².

Take the specific weight of air as 0.0116 kN/m³.

[12M]

5. Fig. shows three reservoirs connected by pipes. Each pipe is 300 mm in diameter and 1500 m long. Assuming co-efficient of friction for each pipe, $f = 0.01$ find the discharge in each pipe.

[12M]



6.A. The velocity distribution in the boundary layer is given by $u/U = (y/\delta)^{1/7}$

Calculate; (i) Displacement thickness, (ii) Momentum thickness, (iii) Shape factor, (iv) Energy thickness

B. Define Mach number and derive an expression of bernoulli's equation for adiabatic and isothermal processes.

[6+6M]

7. Define HGL and TEL, Write equations for Major Energy loss, sudden contraction, expansion, pipes in parallel and series. And derive an expression for flow of viscous fluid in two parallel pipes one is at rest and one is fixed in position. [12M]